Jinxin Zhou

Senior Researcher | Microsoft

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Education

The Ohio State University — *Columbus, USA*
Ph.D. in Computer Science and Engineering — *2022 – 2026*
Advisor: Prof. Zhihui Zhu

University of Michigan, Ann Arbor — *Ann Arbor, USA*
M.S. in Electrical Engineering and Computer Science — *2019 – 2021*

Huazhong University of Science and Technology — *Wuhan, China*
B.S. in Electrical Engineering and Automation — *2015 – 2019*

Work Experience

Feb. 2026 – Present	**Senior Researcher**, Microsoft	*Redmond, USA*
May 2024 – Aug. 2024	**Machine Learning Engineer Intern**, Adobe	*San Jose, USA*
May 2023 – Aug. 2023	**Research Intern**, Microsoft	*Redmond, USA*
Aug. 2022 – Jan. 2026	**Research Assistant**, The Ohio State University	*Columbus, USA*
Jan. 2020 – Dec. 2020	**Research Assistant**, University of Michigan	*Ann Arbor, USA*

Publications

** indicates equal contribution.*

Published

1. **Jinxin Zhou**, Jiachen Jiang, and Zhihui Zhu, "Improving Visual Discriminability of CLIP for Training-Free Open-Vocabulary Semantic Segmentation". *Transactions on Machine Learning Research (**TMLR**),* 2026.

2. Zhen Qin*, **Jinxin Zhou***, Jiachen Jiang, and Zhihui Zhu, "On the Convergence of Gradient Descent on Learning Transformers with Residual Connections". *IEEE Signal Processing Letters (**IEEE SPL**),* 2026.

3. Jiachen Jiang, **Jinxin Zhou**, Bo Peng, Xia Ning, and Zhihui Zhu, "Analyzing Fine-Grained Alignment and Enhancing Vision Understanding in Multimodal Language Models". *Neural Information Processing Systems (**NeurIPS**),* 2025.

4. Jiachen Jiang, **Jinxin Zhou**, and Zhihui Zhu, "On Layer-wise Representation Similarity: Application for Multi-Exit Models with a Single Classifier". *International Conference on Learning Representations (**ICLR**),* 2025.

5. Lijun Ding, Zhen Qin, Liwei Jiang, **Jinxin Zhou**, and Zhihui Zhu, "A Validation Approach to Over-parameterized Matrix and Image Recovery". *Conference on Parsimony and Learning (**CPAL**),* 2025.

6. **Jinxin Zhou**, Jiachen Jiang, and Zhihui Zhu, "Are All Layers Created Equal: A Neural Collapse Perspective". *Conference on Parsimony and Learning (**CPAL**),* 2025.

7. **Jinxin Zhou***, Tianyu Ding*, Tianyi Chen, Jiachen Jiang, Ilya Zharkov, Zhihui Zhu, and Luming Liang, "DREAM: Diffusion Rectification and Estimation-Adaptive Models". *Computer Vision and Pattern Recognition (**CVPR**),* 2024.

8. Jiachen Jiang*, **Jinxin Zhou***, Peng Wang, Qing Qu, Dustin Mixon, Chong You, and Zhihui Zhu, "Generalized Neural Collapse for A Large Number of Classes". *International Conference on Machine Learning (**ICML**),* 2024.

9. Xiao Li, Sheng Liu, **Jinxin Zhou**, Xinyu Lu, Carlos Fernandez-Granda, Zhihui Zhu, and Qing Qu, "Principled and Efficient Transfer Learning of Deep Models via Neural Collapse". *Transactions on Machine Learning Research (**TMLR**),* 2023.
10. **Jinxin Zhou**, Chong You, Xiao Li, Kangning Liu, Sheng Liu, Qing Qu, and Zhihui Zhu, "Are All Losses Created Equal: A Neural Collapse Perspective". *Neural Information Processing Systems (**NeurIPS**),* 2022.
11. **Jinxin Zhou***, Xiao Li*, Tianyu Ding, Chong You, Qing Qu, and Zhihui Zhu, "On the Optimization Landscape of Neural Collapse under MSE Loss: Global Optimality with Unconstrained Features". *International Conference on Machine Learning (**ICML**),* 2022.
12. Zhihui Zhu*, Tianyu Ding*, **Jinxin Zhou**, Xiao Li, Chong You, Jeremias Sulam, and Qing Qu, "A Geometric Analysis of Neural Collapse with Unconstrained Features (**Spotlight, top 3%**)". *Neural Information Processing Systems (**NeurIPS**),* 2021.
13. **Jinxin Zhou**, Chuan Zhou, Heang-Ping Chan, Lubomir M. Hadjiiski, and Qian Dong, "Deep Learning Based Similarity-Consistency Abnormality Detection (SCAD) Model for Classification of MRI Patterns of Multiple Myeloma Infiltration". *Proceedings of SPIE Conference on Medical Imaging (**SPIE**),* 2021.

Preprints

1. Jiachen Jiang, Yuxin Dong, **Jinxin Zhou**, and Zhihui Zhu, "From Compression to Expansion: A Layerwise Analysis of In-Context Learning". *arXiv:2505.17322,* 2025.
2. Yong Ma, Luming Liang, Tianyu Ding, **Jinxin Zhou**, "D-STAR: Diffusion Model with Spatial-Temporal Adaptive Rectification". *arXiv preprint,* 2025.
3. Jiachen Jiang, Tianyu Ding, Ke Zhang, **Jinxin Zhou**, Tianyi Chen, Ilya Zharkov, Zhihui Zhu, and Luming Liang, "Cat-AIR: Content and Task-Aware All-in-One Image Restoration". *arXiv:2503.17915,* 2025.
4. Tianyu Ding, Tianyi Chen, Haidong Zhu, Jiachen Jiang, Yiqi Zhong, **Jinxin Zhou**, Guangzhi Wang, Zhihui Zhu, Ilya Zharkov, and Luming Liang, "The Efficiency Spectrum of Large Language Models: An Algorithmic Survey". *arXiv:2312.00678,* 2023.
5. Tianyu Ding, **Jinxin Zhou**, Tianyi Chen, Zhihui Zhu, Ilya Zharkov, and Luming Liang, "AdaContour: Adaptive Contour Descriptor with Hierarchical Representation". *arXiv:2404.08292,* 2023.

Patents

1. Tianyu Ding, **Jinxin Zhou**, Tianyi Chen, Ilya Zharkov, and Luming Liang, "Adaptive Model for Super-Resolution". *Filed through Microsoft, **US Patent 12,586,155**,* 2026.
2. **Jinxin Zhou**, Hui Qu, Saeid Mottian, Krishna Kumar Singh, and Ratheesh Kalarot, "Boximotion: Still Images into Dynamic Videos with Bounding Box Control". *Filed through Adobe, in progress,* 2024.

Professional Service

Conference Reviewer

Computer Vision and Pattern Recognition *(2024, 2025)*
International Conference on Computer Vision *(TBD; if applicable)*
International Conference on Learning Representations *(2022, 2023, 2024, 2025)*
International Conference on Machine Learning *(2022, 2023, 2024, 2025)*
Neural Information Processing Systems *(2023, 2024, 2025)*
IEEE International Conference on Acoustics, Speech and Signal Processing *(2023)*

Journal Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence ***(TPAMI)***
Journal of Machine Learning Research ***(JMLR)***
Transactions on Machine Learning Research ***(TMLR)***